Siddharth Saminathan

AI Product Engineer with 4+ years building and shipping production AI systems that automate mission-critical enterprise

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AI Product Engineer

Programming · AI Systems & Agentic Architectures · Backend & Real-Time Systems

Python | SQL | LLM APIs | RAG pipelines | FastAPI

FOCUS AREAS

- Python
- SQL
- AI Product Systems
- Production AI
- Real-time Infrastructure
- API Design

CONTACT

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EDUCATION

M.Sc. Statistics and Machine Learning
Linköping University, 2020-2022

B.Tech Computer Science and Engineering
SRM University, 2016-2020

SKILLS

Programming
Python, SQL

AI Systems
LLM APIs, RAG pipelines, embeddings, prompt engineering, tool calling

Backend
FastAPI, Redis (caching, queues), PostgreSQL, REST APIs, async pipelines

Systems Design
Stateful systems, workflow orchestration, memory systems, context modeling, distributed pipelines

Infra
Docker, AWS, Fly.io, logging, tracing

Data
ETL pipelines, data validation, Power BI, Tableau

LANGUAGES

English (Fluent)
Tamil
Hindi

THESIS

Developed a machine learning pipeline utilizing autoencoders and classifiers to predict colorectal cancer metastasis from RNA sequencing data. Employed transfer learning, PCA, and network analysis for...

PROFILE

AI Product Engineer with 4+ years building and shipping production AI systems that automate mission-critical enterprise workflows, from AI-driven quality management (DFMEA. PFMEA, Control Plans, 8D) to real-time AI products at scale. I take systems from concept to production and optimize them relentlessly: latency, cost, reliability, and user outcomes. Multi-agent orchestration, tool-driven architectures, and memory systems. What I build doesn't just respond, it improves over time.

EXPERIENCE

AI Engineer
Omnex Systems, Chennai *Jul 2025 - Present*

Omnex is an **Enterprise AI Quality Platform (EQMS)** serving global manufacturing. I built their agentic AI system for end-to-end manufacturing quality digitalization, automating the core quality engineering workflows that automotive, aerospace, and industrial OEMs run on.

- Built a production multi-agent AI system that automates end-to-end manufacturing quality workflows: DFMEA (Design Failure Mode), PFMEA (Process Failure Mode), Control Plans, and 8D corrective action, reducing manual engineering effort and improving consistency across global supply chains.
- Designed a tool-driven orchestration layer that dynamically selects tools, routes data across internal and external enterprise systems, and executes multi-step quality workflows (generate to validate to update to act) with full traceability.
- Enabled context-aware quality decisioning: the system maps inputs to production hierarchies (part/process/plant), retrieves relevant historical quality cases, and suggests structured corrective actions and process updates.
- Improved system performance and reliability, reduced workflow latency (~12.5s to ~8s), built validation layers and retry logic, and instrumented full observability (logging, tracing, metrics) for production manufacturing environments.

Co-Founder / AI Engineer
Emote (emotenow.app) *Apr 2024 - Present*

- Built and iterated a production AI system that learns from user interactions over time, modeling context, relationships, and behavioral patterns across sessions to improve response quality and continuity
- Designed and scaled the end-to-end backend architecture (FastAPI, Redis, PostgreSQL, real-time pipelines, multi-model routing), enabling low-latency, stateful AI interactions in production
- Drove core system performance and efficiency, reducing latency (~15s to ~3s) and Inference cost (~$1.20 to ~$0.023) through pipeline optimization, caching, and model selection
- Owned the full product loop (0 to 1 to iteration): scaled to 450+ users with 40+ power users, improved activation (20% to 75%) and retention (5% to 13-15%) by continuously shipping improvements based on real user behavior and feedback.

Data Engineer
Akkodis AB, Sweden *Nov 2022 - Nov 2024*

- Led migration from Azure to PostgreSQL, automating ETL processes using Docker, Argo, and Python.
- Developed and optimized Cron jobs to enhance workflow efficiency.
- Improved Power BI dashboards and metrics pipelines, resulting in 5% increase in business reporting accuracy.
- Constructed automated validation layers to ensure data quality and integrity across enterprise reporting systems.
- Enhanced ETL workflows through Docker containerization and PostgreSQL optimization.

Data Analyst
Positive Integers Pvt Ltd, Chennai *Sep 2019 - May 2020*

- Analyzed financial datasets related to loan portfolios, credit risk, and customer behavior patterns for an NBFC
- Developed interactive dashboards using Tableau and Grafana to visualize business KPIs and financial metrics
- Constructed SQL-based ETL workflows.

KEY ACHIEVEMENTS

- **Built Emote from zero:** Took product from vision to production, grew to 450+ users, achieved 13-15% weekly retention through continuous iteration.
- **Latency reduction:** Drove inference latency from ~15s to ~3s and cost from ~$1.20 to ~$0.023 per session at Emote.
- **Enterprise migration:** Led Azure to PostgreSQL data platform migration at Akkodis for Ericsson systems; recognized for improving reporting reliability and delivery.
- **Agentic manufacturing quality system:** Built production multi-agent AI for enterprise quality management (EQMS) at Omnex Systems, automates DFMEA, PFMEA, Control Plans, and 8D across global manufacturing supply chains (12.5s to 8s latency).